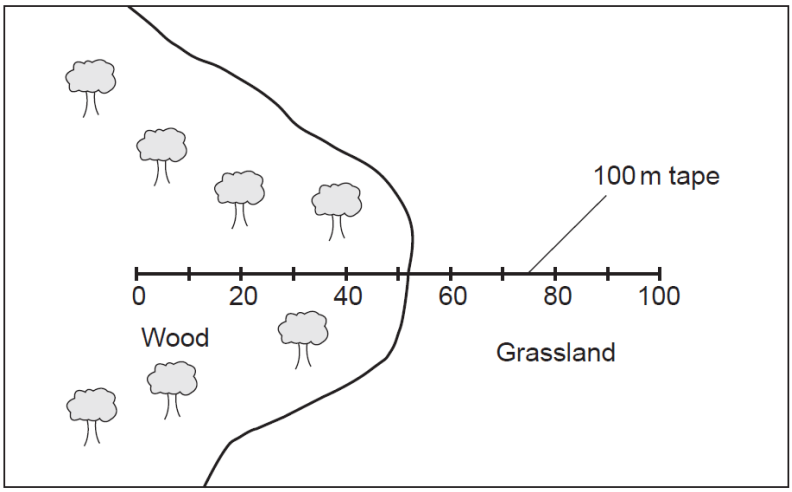
Short questions

|!|BCESB0200800001|!| ★

CCEA GCE AS Biology Assessment Unit AS 3 May 2018 Q2

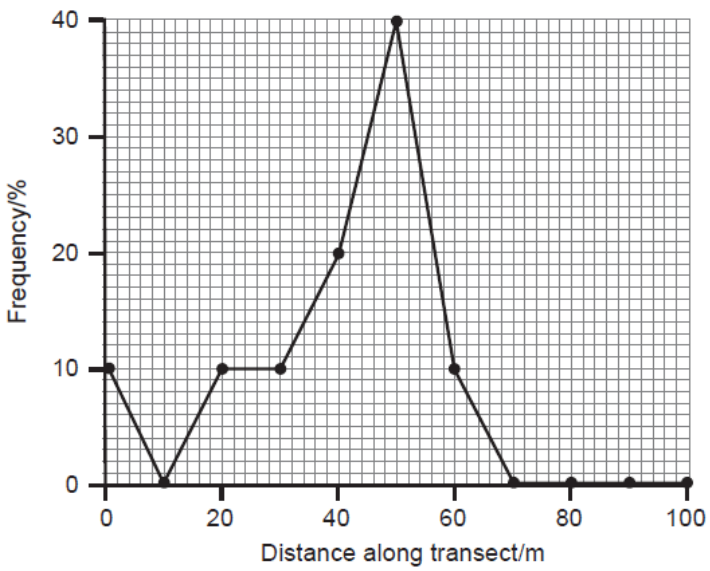
In ecological sampling, species abundance can be measured in a variety of ways. In one investigation, a 100 m line tape was laid out, extending from the centre of a small wood into an area of grassland, as shown in the diagram below.



The frequency of lesser celandine (*Ranunculus ficaria*), a small ground layer plant, was sampled at 10 m intervals along the transect using a pin frame.

A pin frame can be used to measure the frequency of plant species along a transect. At each sampling point, the ten pins in the frame were lowered, and the number of pins which touched lesser celandine was recorded. This data was used to estimate percentage frequency.

The results are shown in the graph below.



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(a) With reference to habitat, describe the distribution shown by the lesser celandine along the transect. (3 marks)

(b) The distribution of lesser celandine is affected by light intensity and competition from other ground layer plants (e.g. grasses). Taking account of this, explain the frequency of lesser celandine at a distance of 50 m along the transect. (1 mark)

##

(a) Any ***three*** of the following: (3)

- In the centre of the wood, the frequency of lesser celandine is low. (1)
- Frequency increases to reach a peak at the edge of the wood. (1)
- Frequency decreases outside the wood into the grassland. (1)
- Between 70–100 m in the grassland, no lesser celandine is present. (1)

(b) There is sufficient light and no or little competitive grasses growing at the wood edge. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	1

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|!|BCESB0200200002|!| ★

The table below shows the types of relationships between organisms.

Type of relationship	Effect on organisms	
	Species X	Species Y
Predation	+	–
Competition		
	+	0
	+	+
Parasitism		

(Remarks: + represents benefited; – represents harmed; 0 represents unaffected)

(a) Complete the column 'Type of relationship' in the above table. (2 marks)

(b) Using the given symbols, state the effect on species X and Y

(i) if they compete for the same resources. (1 mark)

(ii) if species X is a parasite living inside species Y. (1 mark)

(c) Considering a predator-prey relationship between species X and Y, how would the population size of species Y change when the population size of species X declined? (1 mark)

##

(a)

Type of relationship	Effect on organisms	
	Species X	Species Y
Predation	+	–
Competition		
Commensalism (1)	+	0
Mutualism (1)	+	+
Parasitism		

(b) (i) Species X: –
Species Y: – (1)

(ii) Species X: +
Species Y: – (1)

(c) The population size of species Y would increase. (1)

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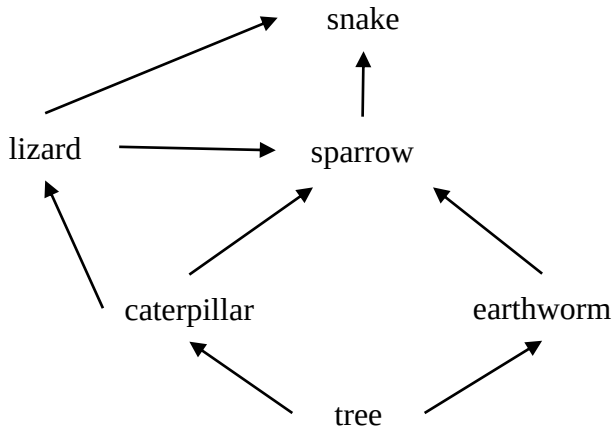
(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)	Applying concepts	1

|!|BCESB0200300003|!| ★

The diagram below shows a food web in a habitat.



(a) Based on the above food web, state a food chain that consists of five trophic levels. (1 mark)

_____

(b) Construct a pyramid of numbers for the food chain you have given in (a). (2 marks)

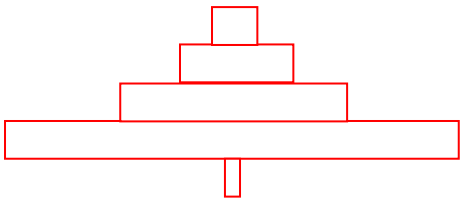
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(c) Explain the shape of this pyramid of numbers. (4 marks)

##

(a) tree → caterpillar → lizard → sparrow → snake (1)

(b)



Correct shape (1)

Correct labels (1)

(c) The number of organisms in each trophic level decreases from caterpillars to snakes. (1)

This is because energy is lost from one trophic level to the next (1)

due to respiration / excretion / incomplete ingestion and digestion of food / death. (1)

The size of a tree is much greater than that of a caterpillar, thus a tree can support many caterpillars. (1)

(7 marks)

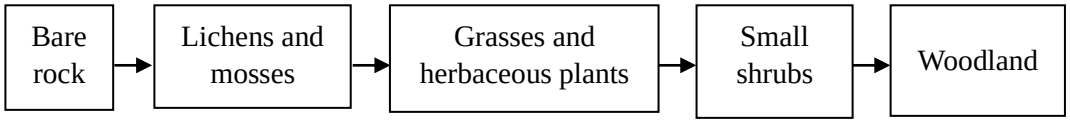
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	1
	Presenting information	1
(c)	Applying concepts	4

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|!|BCESB0200200004|!| ★

The flowchart below shows a series of changes during primary succession on a bare rock.



(a) (i) Identify the pioneer community shown in the flowchart. (1 mark)

_____

(ii) List ***two*** functions of pioneer species during primary succession. (2 marks)

_____

_____

(b) Explain why the species composition in a habitat changes during primary succession. (2 marks)

_____

_____

_____

(c) A hill fire occurs and destroys a woodland habitat.
Explain why the subsequent secondary succession in the destroyed woodland is usually faster than primary succession. (1 mark)

_____

_____

##

(a) (i) Lichens and mosses (1)

(ii) Any ***two*** of the following: (2)

- Modify habitats to support other species
- Form soil
- Improve water retention in the soil

(Or other reasonable answers)

(b) Colonizing species are constantly altering their habitat's environment. (1)
Species better suited to the changing environment may become dominant, while those that perform less well are eliminated. (1)

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(c) Soil already exists / Seeds, roots and underground vegetative parts of plants are still present in the habitat. (1)

(6 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Understanding basic concepts	2
(b)	Applying concepts	2
(c)	Understanding basic concepts	1

|!|BCESB0200200005|!| ★

The following photograph shows a honey bee collecting nectar from a flower.



(a) Identify the ecological relationship between the honey bee and the flowering plant. (1 mark)

(b) Explain your answer to (a). (2 marks)

##

(a) Mutualism (1)

(b) The honey bee gains benefit from the flowering plant by receiving nectar as food. (1)

The flowering plant gains benefit from the honey bee by being pollinated by the latter. (1)

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(3 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2

|!|BCESC0200200006|!|

The following photograph shows a forested area that has been burnt by a hill fire.



(a) How does the hill fire affect the biodiversity of the area? (1 mark)

(b) Which type of succession will now occur in this area? (1 mark)

(c) Describe the process of the succession given in the answer to (b). (4 marks)

##

(a) It reduces the biodiversity of the area. (1)

(b) Secondary succession (1)

(c) Organisms are driven away or killed by the hill fire, leaving behind the soil and the dead remains of plants. (1)

Grass and herbs start to grow in the burnt area and form the pioneer community. (1)

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Some pioneer plants die and are replaced by other species. (1)

The forest (the climax community) is re-established a few years after the fire. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	1
(c)	Understanding basic concepts	4

|!|BCESB0200300007|!| ★

The following diagram shows a food chain that illustrates the feeding relationship between three organisms.

tree → worm → bird

(a) Which organism is the producer? Describe the organism's role as a producer in the food chain. (3 marks)

(b) Which organism is the secondary consumer? How should this consumer be categorized based on its diet? (2 marks)

(c) The food chain shown above omits an important group of organisms that are also present in the habitat. Name this group of organisms. (1 mark)

##

(a) The tree is the producer. (1)

The tree captures light energy and converts it into chemical energy that is stored in organic compounds. (1)

The organic compounds can be used as food by the worms. (1)

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(b) The bird is the secondary consumer. (1)

It is a carnivore. (1)

(c) Decomposers (1)

(6 marks)

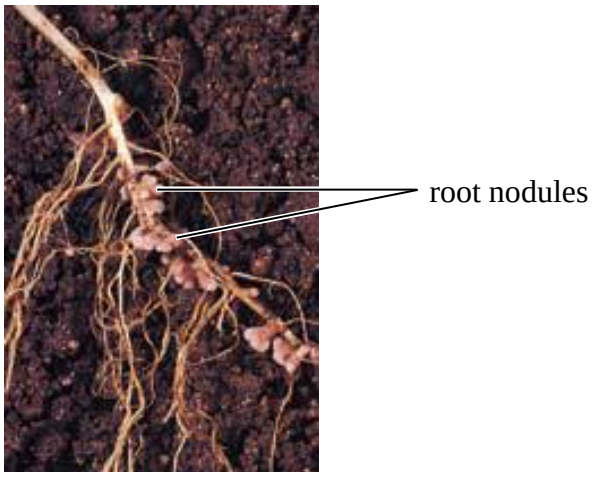
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	2
(c)	Understanding basic concepts	1

|!|BCESC0200500008|!|

✂

The following photograph shows some root nodules of a leguminous plant. Nitrogen-fixing bacteria live inside these root nodules.



(a) Identify the ecological relationship between the nitrogen-fixing bacteria and the leguminous plant. (1 mark)

(b) Describe how nitrogen-fixing bacteria can support the growth of the leguminous plant in infertile soil. (3 marks)

##

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(a) Mutualism (1)

(b) Nitrogen-fixing bacteria fix nitrogen in the soil gas into ammonium compounds. (1)

The leguminous plant uses the ammonium compounds to synthesize proteins. (1)

Proteins can be used to form new cells during growth. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	3

|!|BCESC0200500009|!|

✂

Complete the following section with the appropriate terms. (6 marks)

All living organisms need nitrogen. Plants obtain nitrogen by absorbing (a)_____ from the soil and use them to make (b)_____. Such nitrogen is ultimately transferred along the food chain to (c)_____.

Certain bacteria called (d) _____ can fix the nitrogen in the soil gas into ammonium compounds within their cells.

(e) _____ and fungi are responsible for the breakdown of organic waste and the remains of dead organisms. During such breakdown, the nitrogen in the organic material is converted into ammonium compounds. The ammonium compounds are then oxidized to nitrites and eventually to nitrates by (f)_____.

##

(a) nitrates (1)

(b) proteins (1)

(c) consumers (1)

(d) nitrogen-fixing bacteria (1)

(e) Putrefying bacteria (1)

(f) nitrifying bacteria (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1

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(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESC0200200010|!|

Match each of the terms listed in ***Column 1*** with the correct example in ***Column 2*** by putting the appropriate letters in the spaces provided. (5 marks)

Column 1			***Column 2***	
predation	(a)	_____	A.	barnacle and limpet on rock
parasitism	(b)	_____	B.	hare and lynx
competition	(c)	_____	C.	epiphyte and tree
commensalism	(d)	_____	D.	alga and fungus in lichens
mutualism	(e)	_____	E.	tapeworm and human

##

(a) B (1)
(b) E (1)
(c) A (1)
(d) C (1)
(e) D (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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